

JAMES ZHANG

Senior Full-Stack AI & Automation Engineer

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SUMMARY

Seasoned Senior Full-Stack AI & Automation Engineer with 10+ years architecting end-to-end AI solutions. Expert in RAG/LLM systems, computer-vision pipelines, and data automation workflows. Proven ability scaling AI applications from prototype to production, mentoring technical teams, and optimizing complex data processing. Core strengths: Python, AI/ML frameworks, React/Node.js, workflow automation.

EXPERIENCE

Senior Full-Stack AI & Automation Engineer

FAW Group Co., Ltd., Changchun, China | 07/2023 – Present

- Architected RAG + LLM assistants for 50+ documents enabling 70% faster retrieval; implemented semantic chunking + vector DB (Pinecone/Qdrant)
- Led computer-vision defect detection improving accuracy from 76% to 94%; deployed real-time inference processing 10K+ images/week
- Orchestrated 15+ production automation workflows using Python + n8n/Make.com/Zapier, reducing manual data entry by 80+ hours/month
- Spearheaded Power BI dashboards and automated reports; built VBA macros reducing monthly reporting from 16 to 2 hours
- Engineered React/Node.js APIs translating ML outputs into operational insights; 200+ daily active users, 99.2% uptime
- Implemented scheduled web-scraping pipelines (BeautifulSoup/Selenium) maintaining real-time datasets over 18 months
- Mentored 4-person team; established code review practices reducing deployment errors by 60%

Full-Stack Developer (AI/Computer Vision)

FAW Group Co., Ltd., Changchun, China | 04/2018 – 06/2023

- Delivered RAG + LLM solutions for 40+ documents cutting search time by 65%; systematically improved model quality
- Built Python AI pipelines for computer-vision defect detection; maintained 95%+ accuracy across update cycles
- Designed Power BI dashboards and Excel reporting; reduced overhead by 75% with VBA automation
- Built React/Node.js APIs supporting 5 internal applications with 99.2% uptime
- Engineered web-scraping pipelines maintaining 16+ data sources for ML training and analytics

Junior Developer / Automation Engineer

FAW Group Co., Ltd., Changchun, China | 09/2015 – 03/2018

- Engineered data extraction automation (Python/BeautifulSoup/Selenium) eliminating 40+ hours/month of manual work
- Developed internal PHP + MySQL task management systems improving team productivity by 25%
- Built ETL pipelines standardizing datasets; reduced data quality issues by 50%
- Automated report generation with validated Excel/BI outputs; implemented data quality gates

Software Engineering Intern

FAW Group Co., Ltd., Changchun, China | 06/2015 – 08/2015

- Developed ETL and data cleaning pipelines using Python for downstream processing
- Prototyped Python services for data transformation, validation, and formatting
- Contributed JavaScript improvements to internal web applications; enhanced user experience
- Documented automation workflows and validated data pipeline outputs for reliability

EDUCATION

Bachelor of Science in Computer Science

Tsinghua University, Beijing, China | 08/2011 – 08/2015

- Specialized in AI, Machine Learning, and Software Engineering
- Core coursework: Data Structures, Algorithms, Computer Vision, Deep Learning
- Capstone: Image processing & object detection using OpenCV
- Self-directed: MERN stack, cloud deployment

SKILLS

AI/ML: RAG, LLM, AI Agent, Chatbot, n8n, Make.com, Zapier

Backend: FastAPI, Flask, Django, ASP.NET, Node.js, Laravel, PHP

Frontend: React, Angular, JavaScript, HTML/CSS, MERN

Databases: PostgreSQL, MySQL, MongoDB, SQL Server, Pinecone, Qdrant

ML Libraries: TensorFlow, Keras, PyTorch, scikit-learn, OpenCV

Data Tools: BeautifulSoup, Scrapy, Selenium, Pandas, SQL

DevOps: Git, Docker, AWS, CI/CD, VS Code, Visual Studio

BI/Office: Power BI, Excel (VBA), Google Suite

Programming: Python, JavaScript, PHP, C#, C++, SQL, Bash

LANGUAGES

Chinese: Native or Bilingual Proficiency | **English:** Full Professional Proficiency